Hall Ticket Number:

MR-21

Code: B315CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Discrete Mathematics

(Common to CSE, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

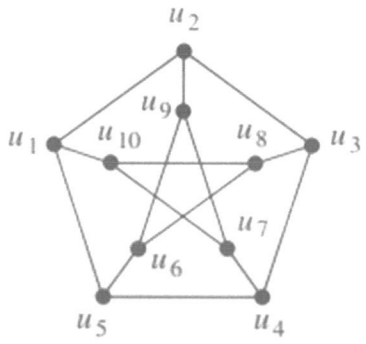
Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Suppose that the domain of the propositional function P(x) consists of the integers −2, −1, 0, 1, and 2. Write out each of these propositions using disjunctions, conjunctions, and negations. i) $\exists x P(x)$ ii) $\forall x P(x)$	2	2	1	1
1. b)	For premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises. "All insects have six legs." "Dragonflies are insects", "Spiders do not have six legs.", "Spiders eat dragonflies".	2	2	1	1
1. c)	Define Equivalence Relation with an example.	2	1	2	1
1. d)	Define Reflexive and Symmetric properties of a relation with examples.	2	1	2	1
1. e)	What is the Co-efficient of xy in $(x+y)^5$	2	1	3	1
1. f)	Find the number of permutations of letters of "Mathematics".	2	1	3	2
1. g)	Write the Axioms of Probability.	2	1	4	1
1. h)	What is Generating Function?	2	1	4	1
1. i)	Define i) planar graph ii) Spanning Tree with Example.	2	1	5	1
1. j)	Draw K_7 and $K_{4,4}$.	2	2	5	2

Part-B (5 x 10 = 50 Marks)

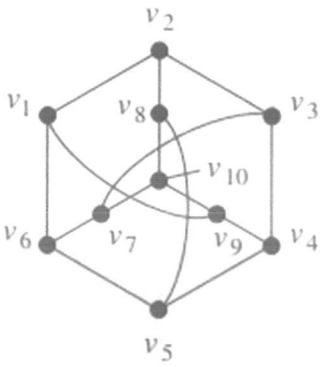
Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Obtain the PDNF and PCNF of $(P \wedge Q) \vee (\neg P \wedge R)$	5	3	1	1
2. b)	Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$ without using truth table.	5	2	1	1
	OR				
2. c)	Show that $r \rightarrow s$ can be derived from the premises $p \rightarrow (q \rightarrow s)$, $\neg r \vee p$ *and* q.	5	2	1	1
2. d)	i) Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction. ii)Use a direct proof to show that the sum of two even integers is even.	5	4	1	2
3. a)	Draw the Hasse diagram representing the partial ordering {(a, b) | a divides b} on{1, 2, 3, 4, 6, 8, 12}.	5	1	2	1
3. b)	Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if $a_n = (-4)^n$?	5	2	2	2
	OR				
3. c)	Draw Hasse Diagram for the poset ({2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72}, |) and answer below questions i) Find the maximal elements and minimal elements. ii) Find all upper bounds and least upper bounds of {2, 9}.	5	1	2	1,2
3. d)	Use Warshall's algorithm to find the transitive closures of the relations in {(2, 1), (2,3), (3,1),(3,4), (4,1), (4, 3)}.	5	1	2	2

P.T.O.

4. a)	Use mathematical induction to show that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n.	5	3	3	1,2
4. b)	How many Permutations can be made with the letters of the word " ENGINEERING".	5	1	3	1
	OR				
4. c)	Show that if n is an integer greater than 1, then n can be written as the product of primes.	5	3	3	1
4. d)	How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?	5	1	3	2
5. a)	State and Prove Bayes Theorem.	5	4	4	1
5. b)	Let f (n) = 5f (n/2) + 3 and f (1) = 7. Find f (2^k), where k is a positive integer. Also, estimate f (n) if f is an increasing function.	5	4	4	1
	OR				
5. c)	Find the solution to the recurrence relation $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.	5	3	4	1,2
5. d)	Find the generating functions for $(1 + x)^{-n}$ and $(1 - x)^{-n}$, where n is a positive integer, using the extended binomial theorem.	5	3	4	1
6. a)	Define isomorphism .Establish an isomorphism for the following the Graphs G and G'	5	3	5	1

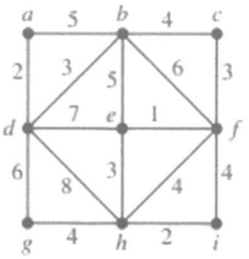


Graph G



Graph G'

6. b)	Explain Tree Traversal Algorithms using Examples.	5	1	5	1
	OR				
6. c)	i) Show that a graph K_n has a Hamiltonian cycle whenever $n \geq 3$. ii) Show that K_5 is an Euler's circuit and also Hamiltonian cycle.	5	1	5	1
6. d)	Use Kruskal's algorithm to find a minimum spanning tree for the weighted graph.	5	3	5	1



M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome